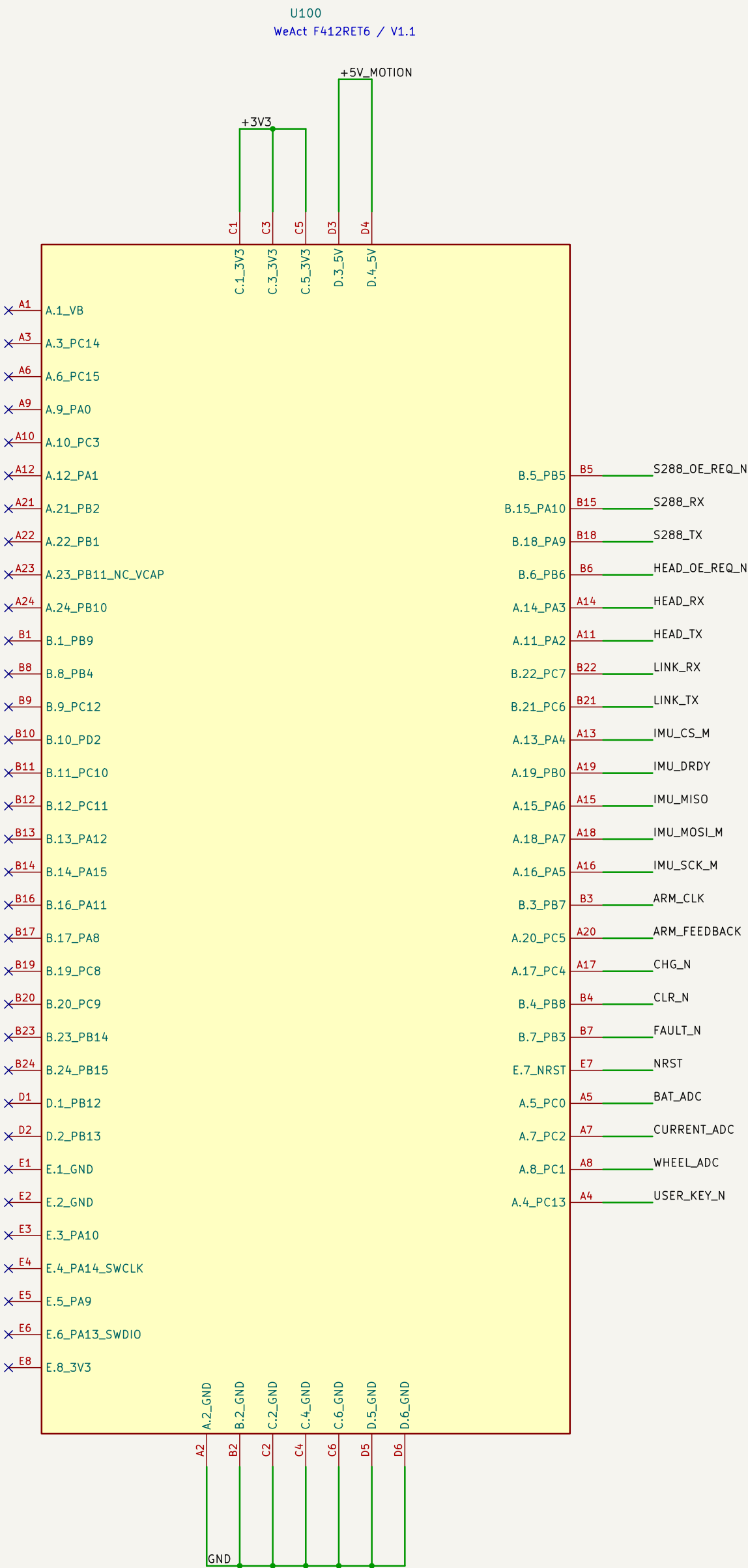
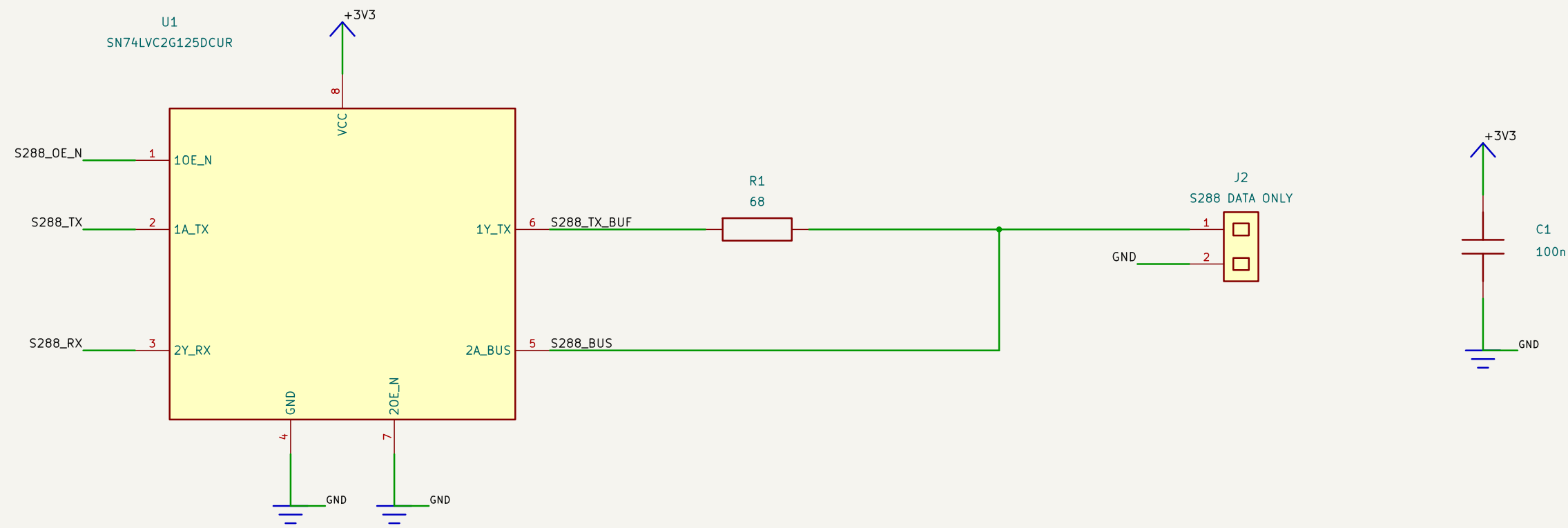


01 STM32 module / GPIO map



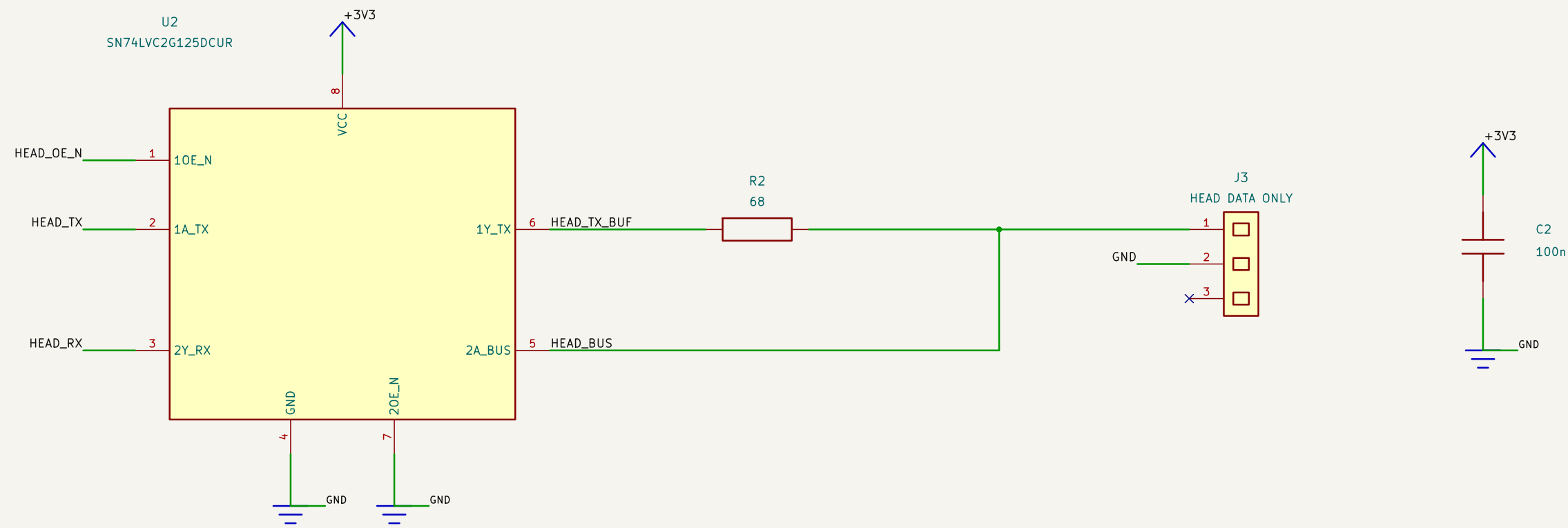
NC pins are intentionally unused. SWD probe connects on the module top.

02 Wheel TTL / 6 Mbps target



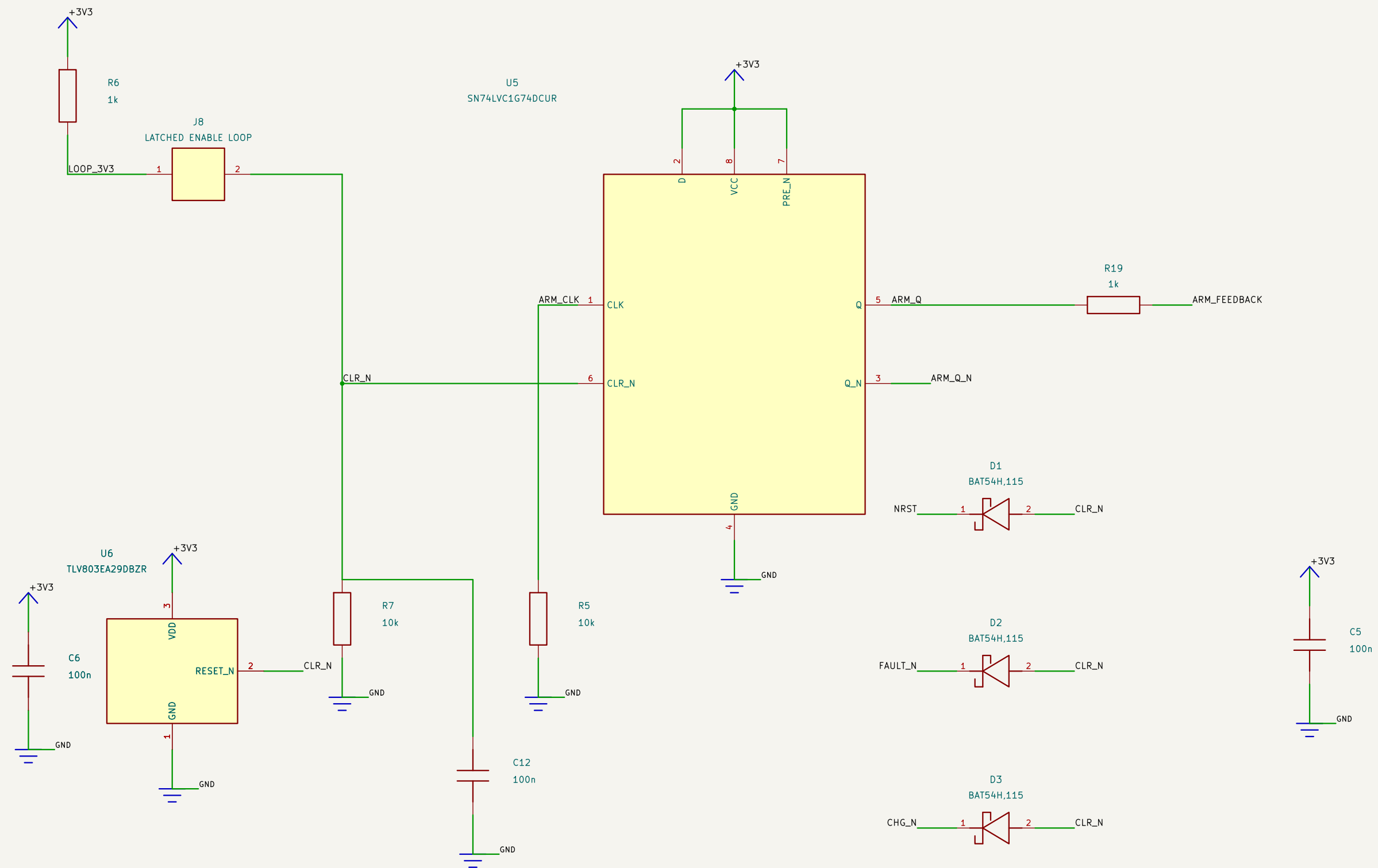
3.3 V logic; actuator thresholds and timing remain bench items.

03 Head TTL / separate bus



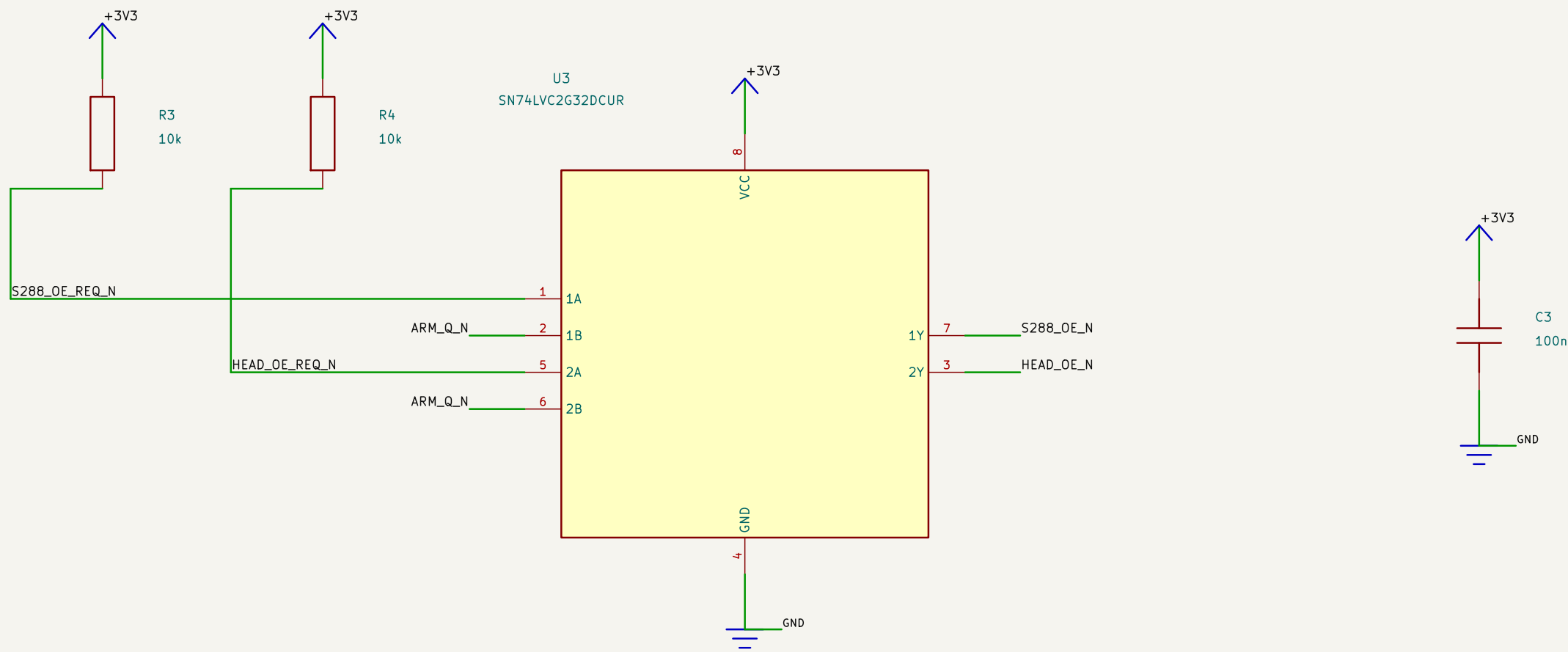
3.3 V logic; actuator thresholds and timing remain bench items.

04 Explicit ARM latch / physical disable loop



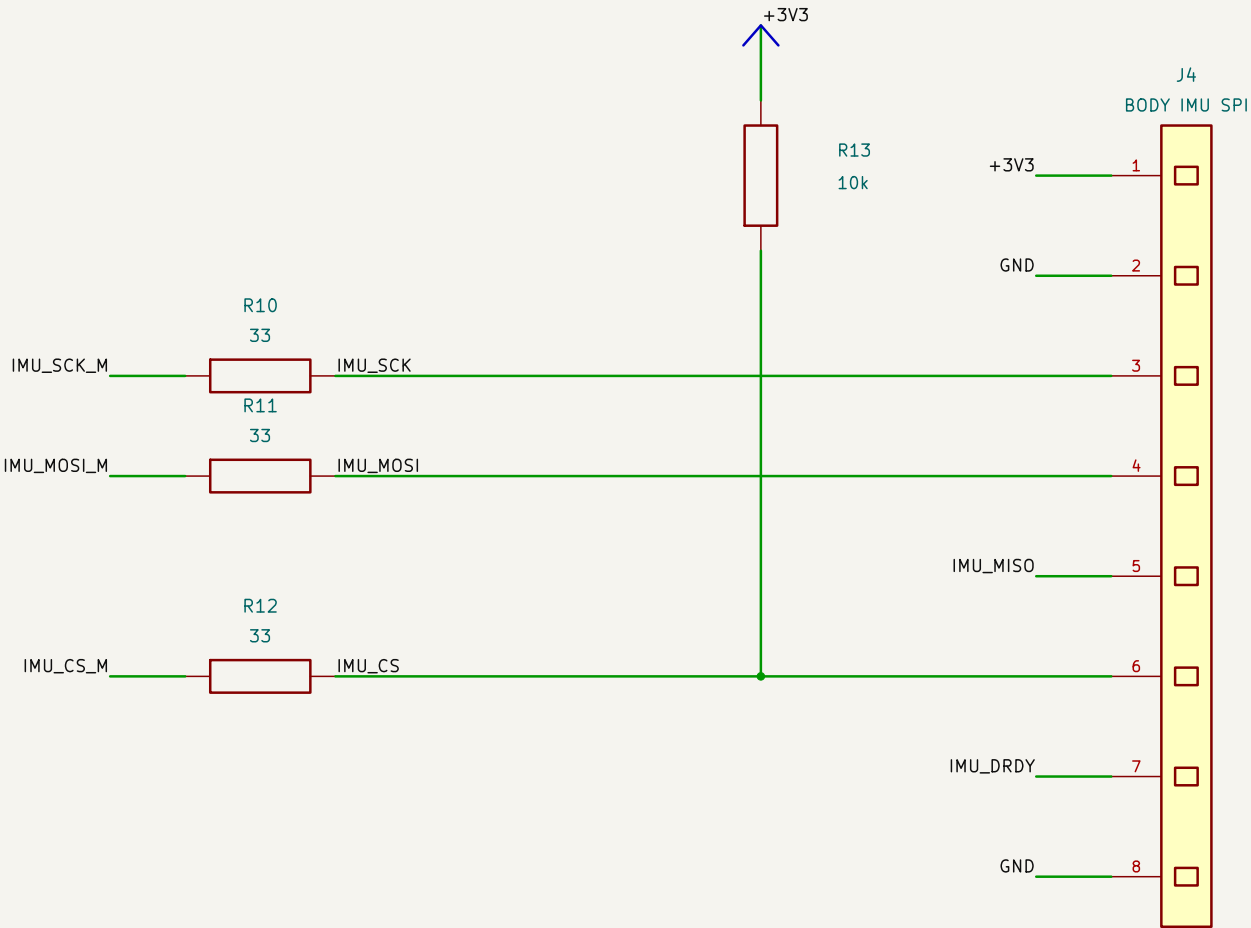
CLR_N dominates. Re-arm requires a new ARM_CLK edge; this is not a watchdog.

05 Transmit-enable gating



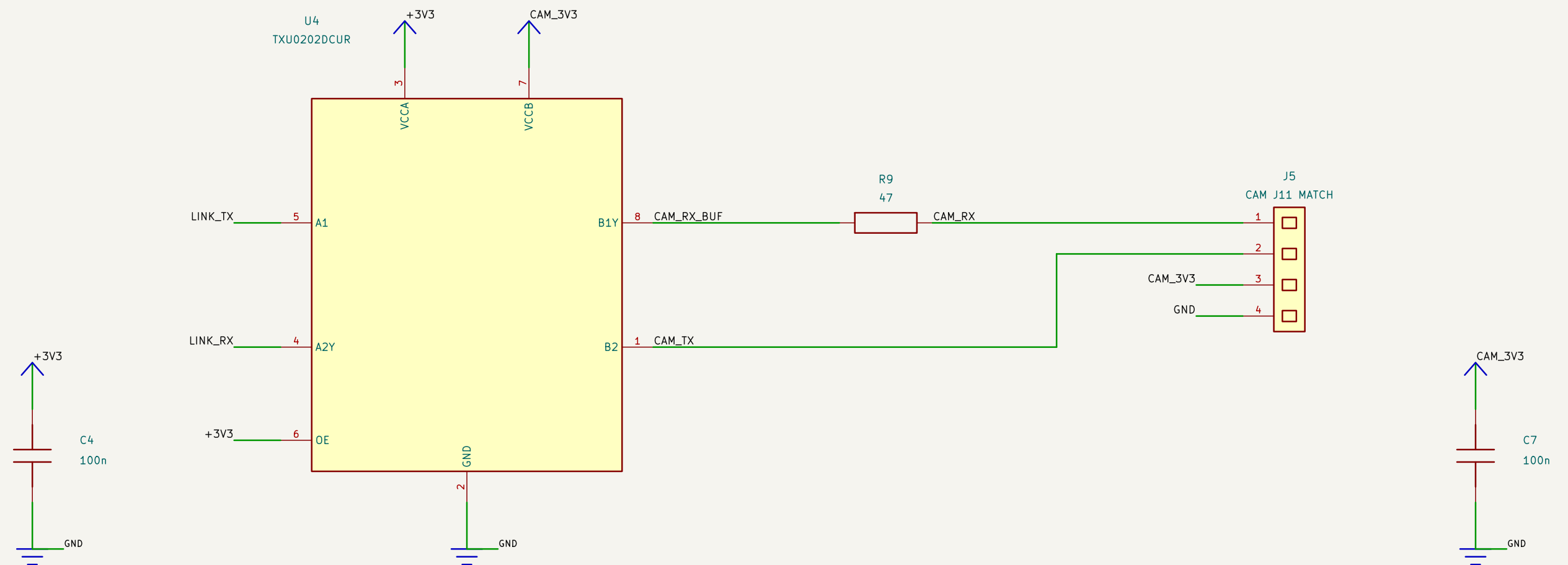
OE_N = request_N OR ARM_Q_N. Pull-ups disable TX by default.

06 Body IMU connector / source damping



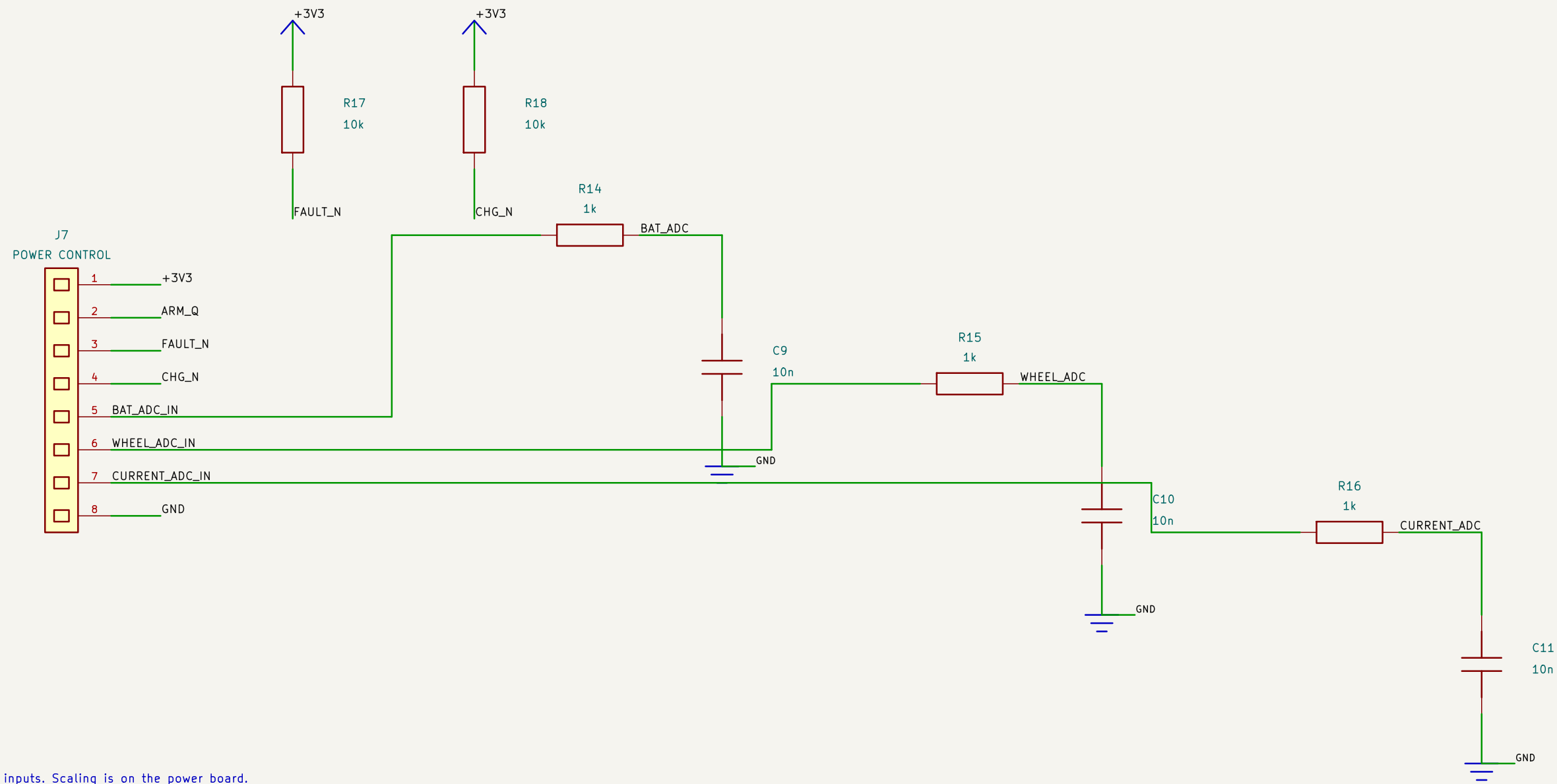
SPI and DRDY belong to the rigid body IMU, not the head.

07 Interaction UART / power-domain isolation



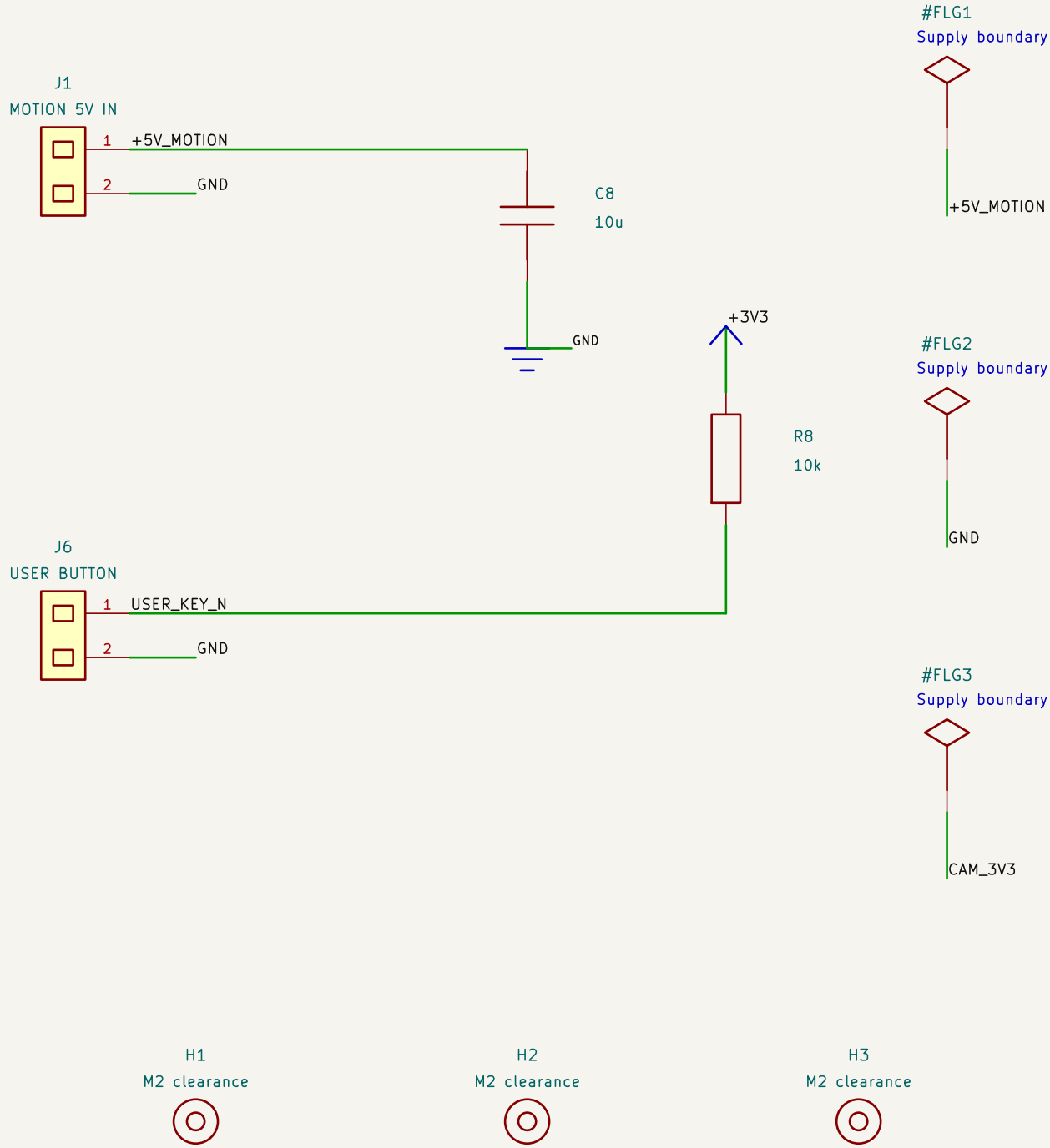
CAM_3V3 powers translator B only. It does not power the motion MCU.

08 Power monitor / RC filters



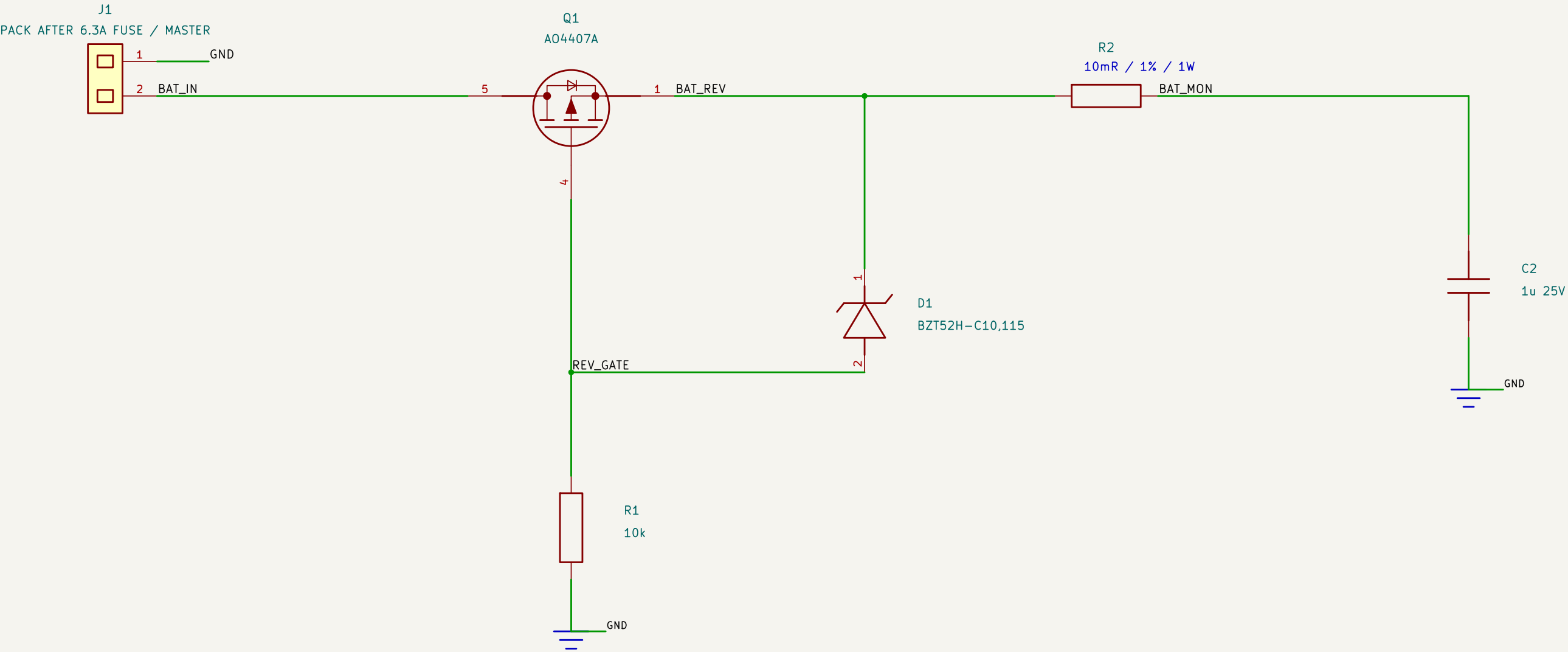
1k / 10nF filters at the ADC inputs. Scaling is on the power board.

09 Input / user button / supply boundaries



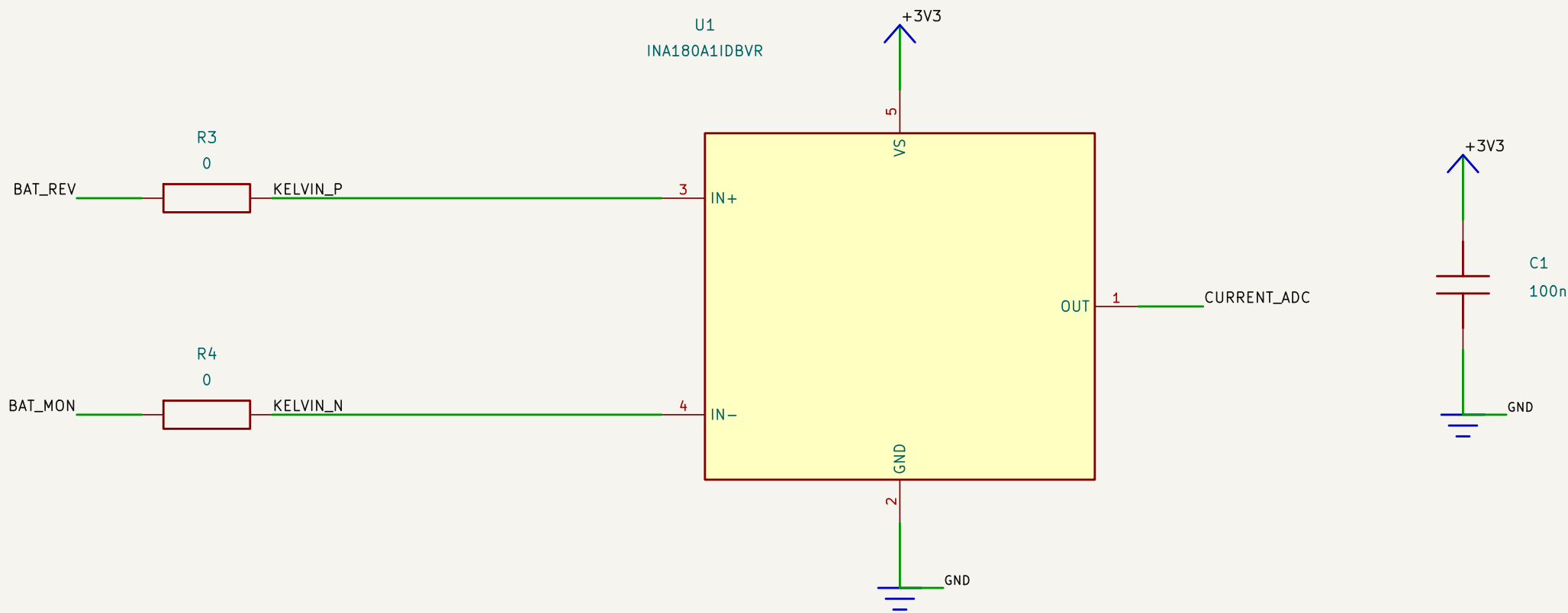
J6 is a user button. Physical disable is J8; reset can cause a fall.

01 Pack input / reverse protection / shunt



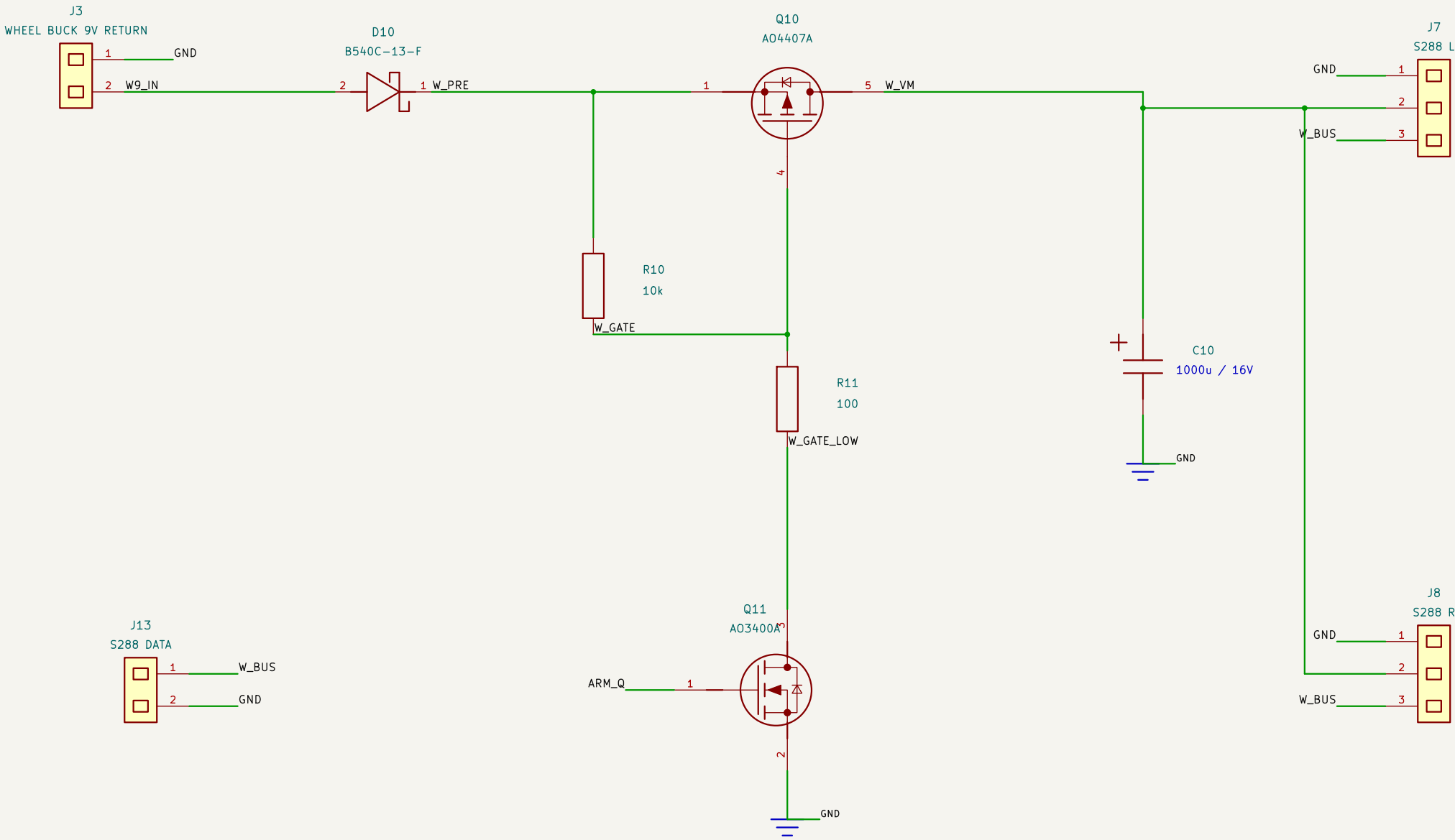
External pack, fuse and master switch precede J1. No charger on this PCB.

02 Kelvin current sense



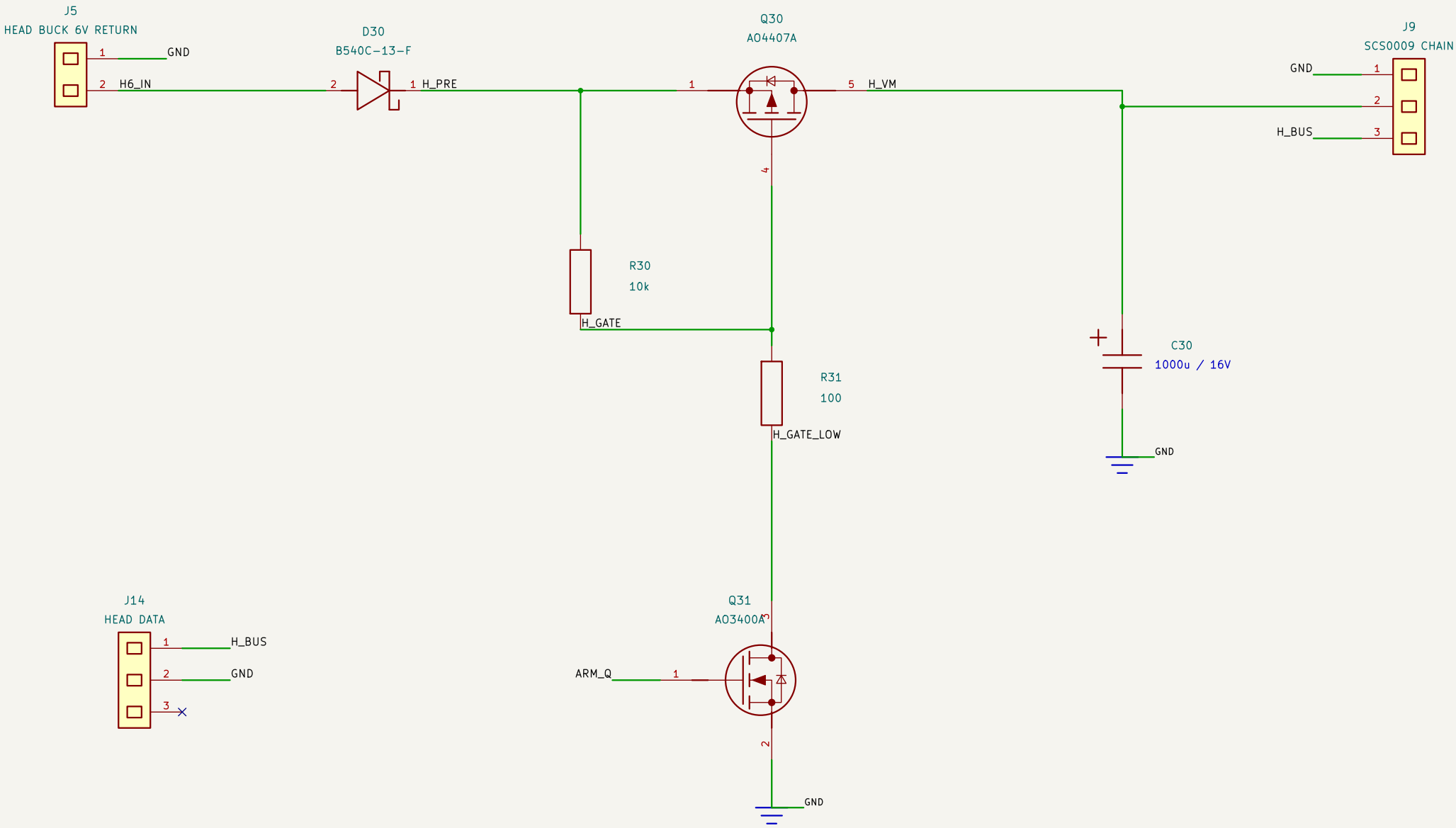
Sense from the shunt pads. Discharge-only nominal transfer: 0.2 V/A.

03 Wheel 9 V conditioning



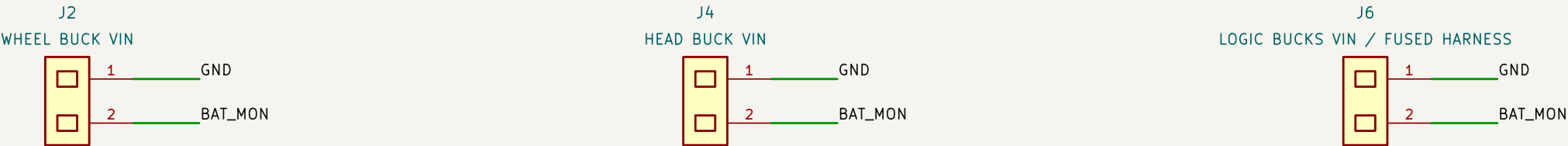
External buck return. Inrush, brake energy and thermal performance: NOT_TESTED.

04 Head 6 V conditioning



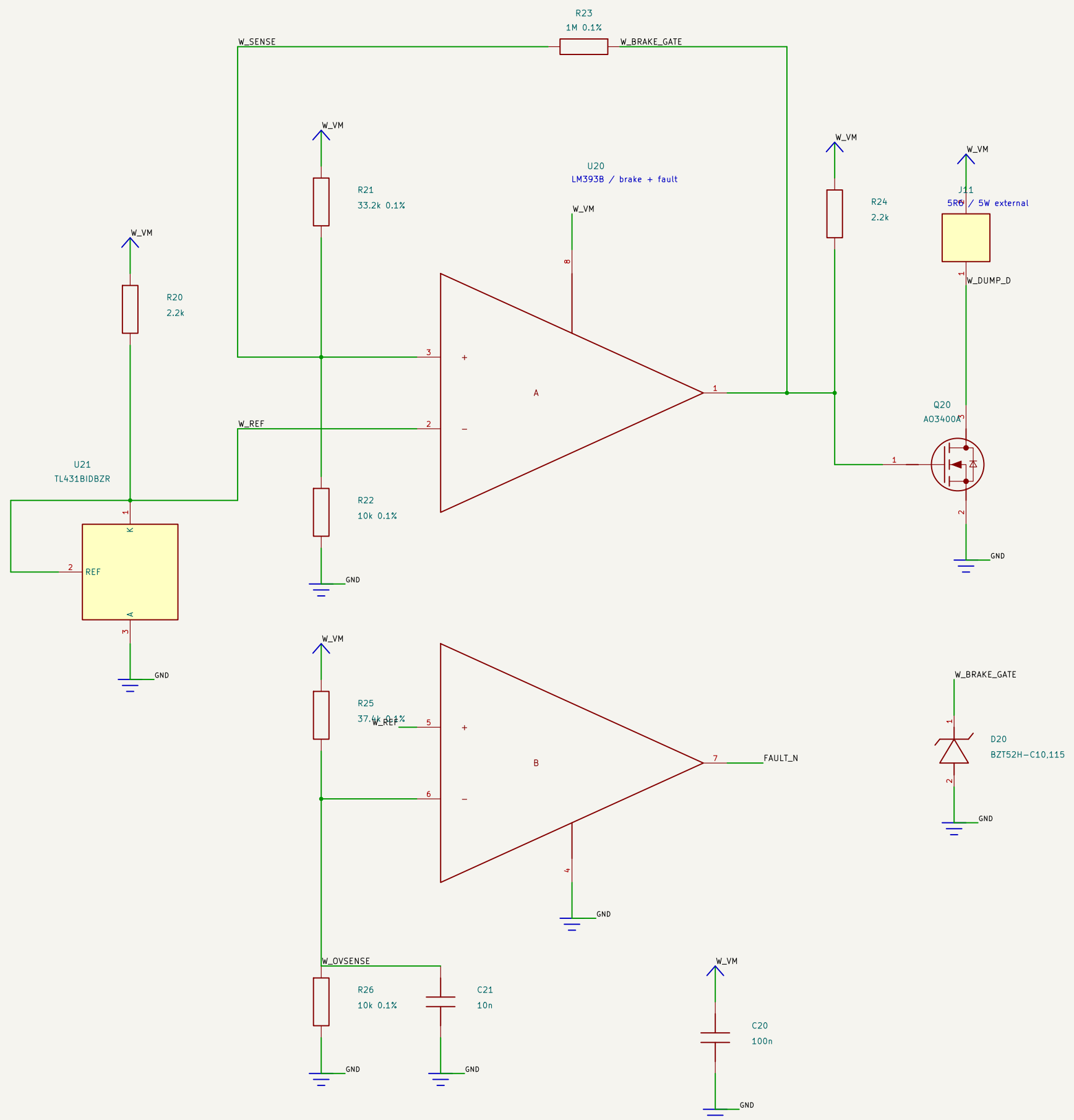
External buck return. Inrush, brake energy and thermal performance: NOT_TESTED.

05 External converter input distribution



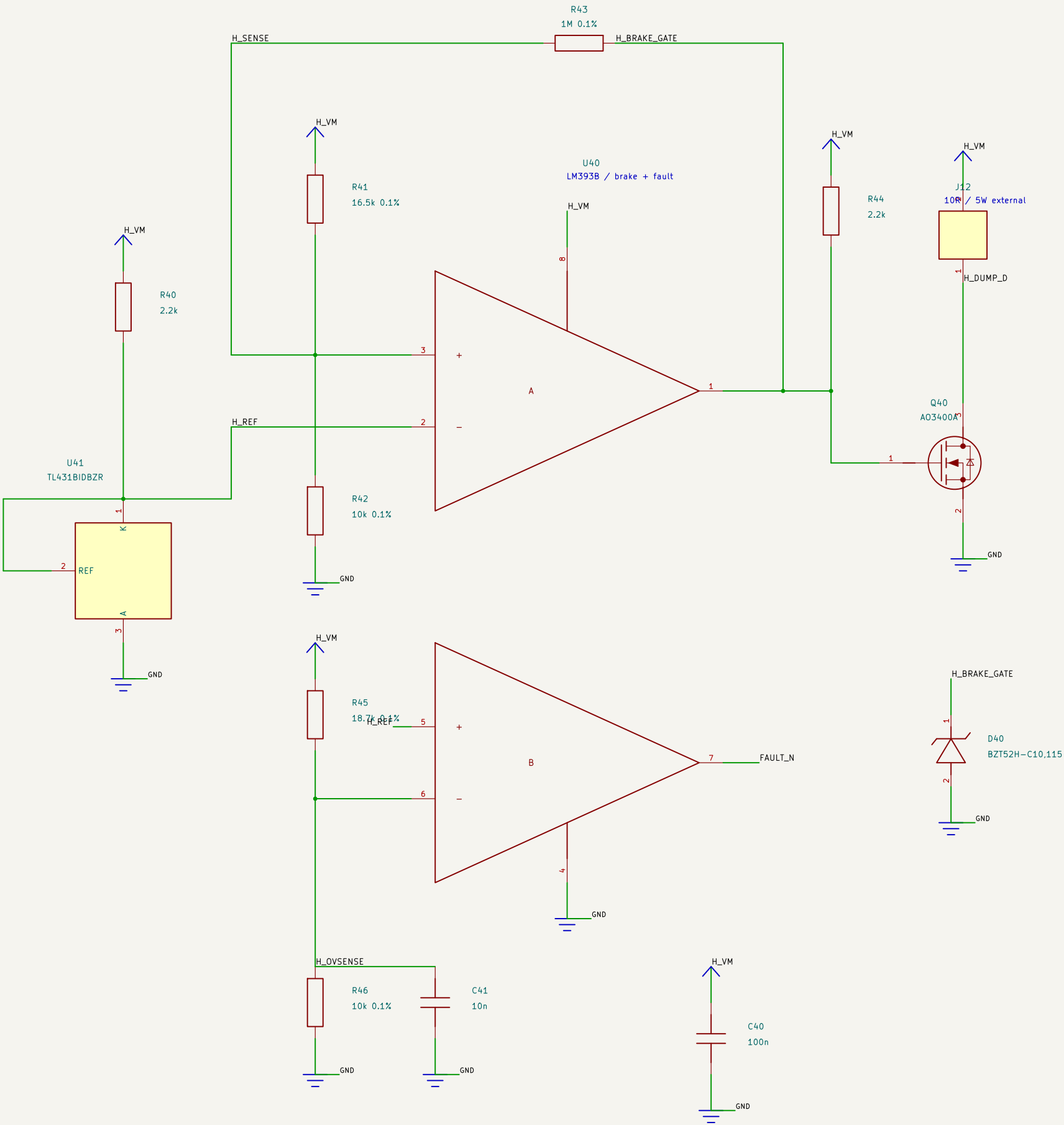
J2 -> wheel buck; J4 -> head buck; J6 -> two independent 5 V converters.

06 Wheel rail / brake and overvoltage

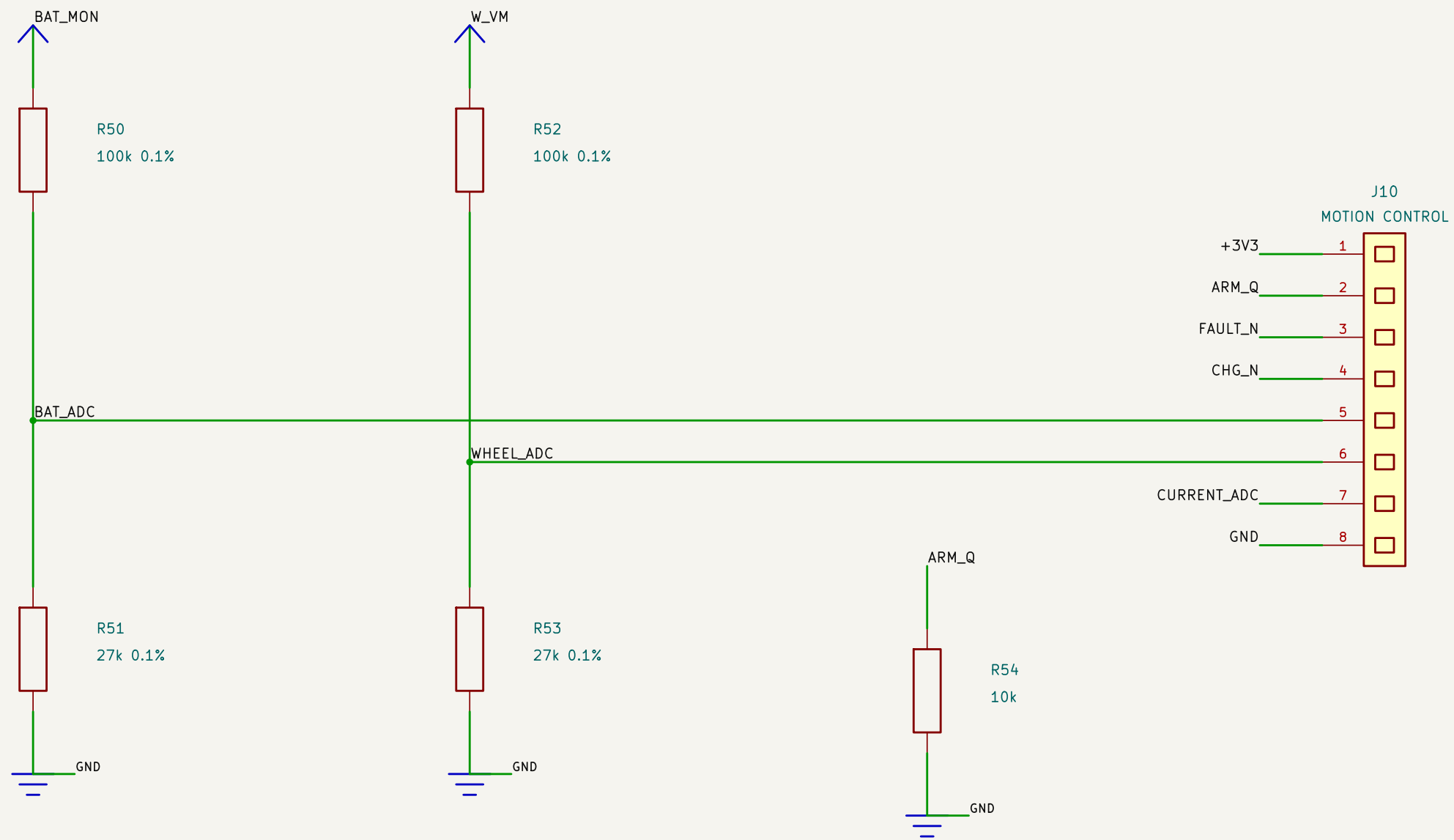


Two comparators share one LM393B. External dump resistor needs pulse qualification.

07 Head rail / brake and overvoltage

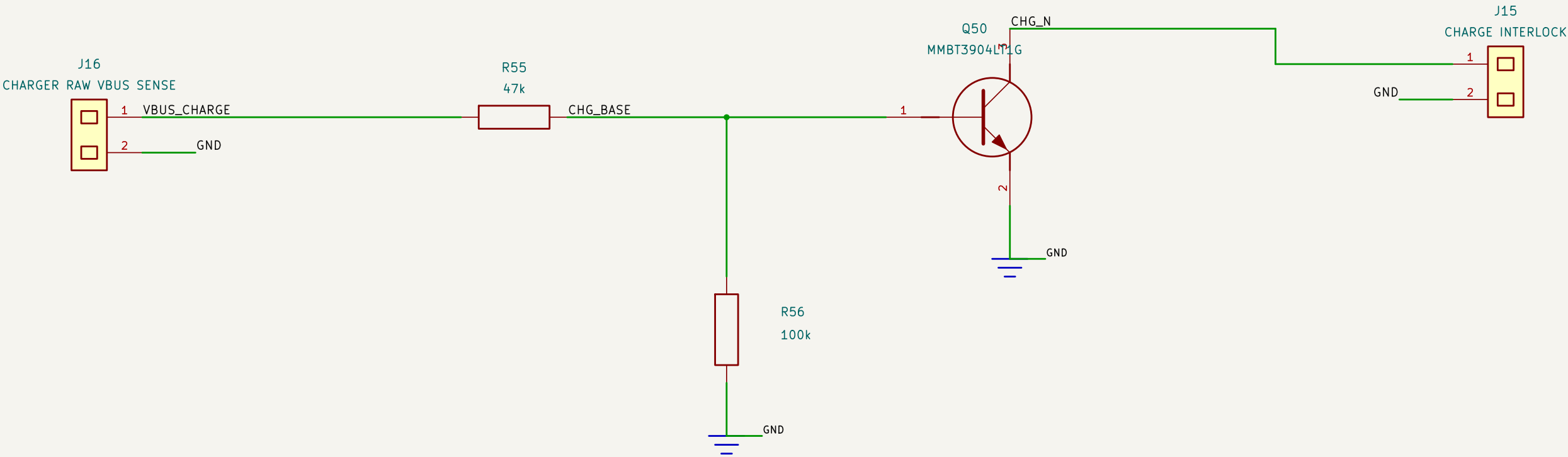


08 Voltage monitor / motion connector



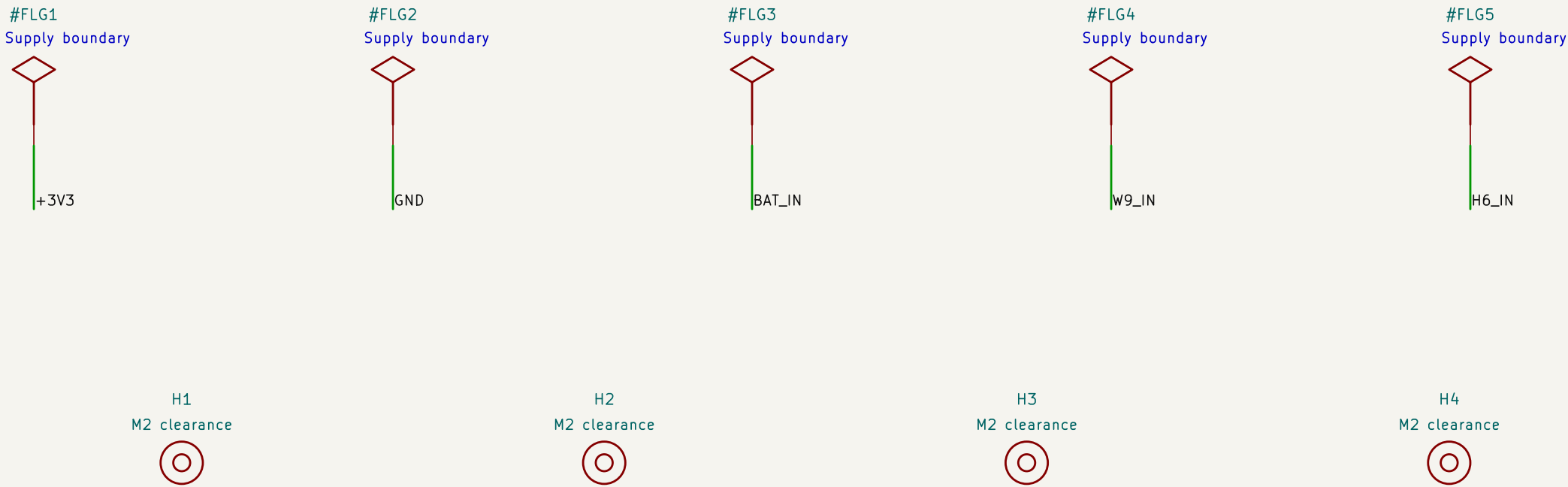
Divider ratio 27 / 127. Current_ADC comes from block 02.

09 Charge insertion / VBUS sense



Sense and interlock only; no CC / PD / CC-CV charger implementation.

10 Supply boundaries / board mounting

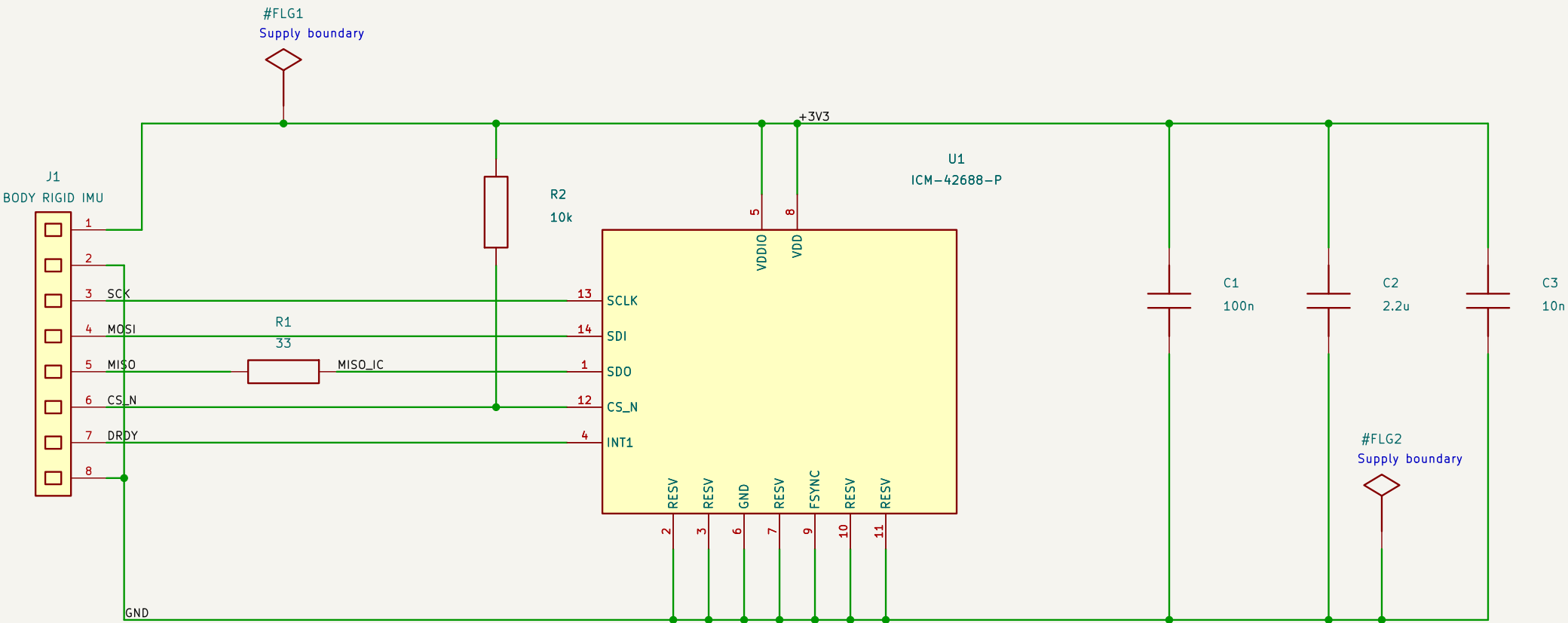


80 x 45 mm PCB; populated assembly and thermal performance are not qualified.

MORI BODY IMU | FUNCTIONAL CIRCUIT

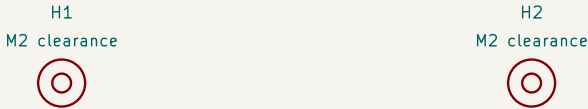
S2 drawing | same pin map and PCB | PROTOTYPE / UNVALIDATED

01 SPI connector / sensor / local decoupling



Body-fixed IMU. Axis transform and assembled pin-1 orientation still require verification.

02 Mechanical references



PROTOTYPE / UNVALIDATED – electrical design unchanged

Sheet: /
File: MORI_imu_P2.kicad_sch

Title: MORI / imu / P1 – FUNCTIONAL SCHEMATIC

Size: A4 Date: 2026-09-22

KiCad E.D.A. 10.0.6

Rev: S2 drawing layout

Id: 1/1